

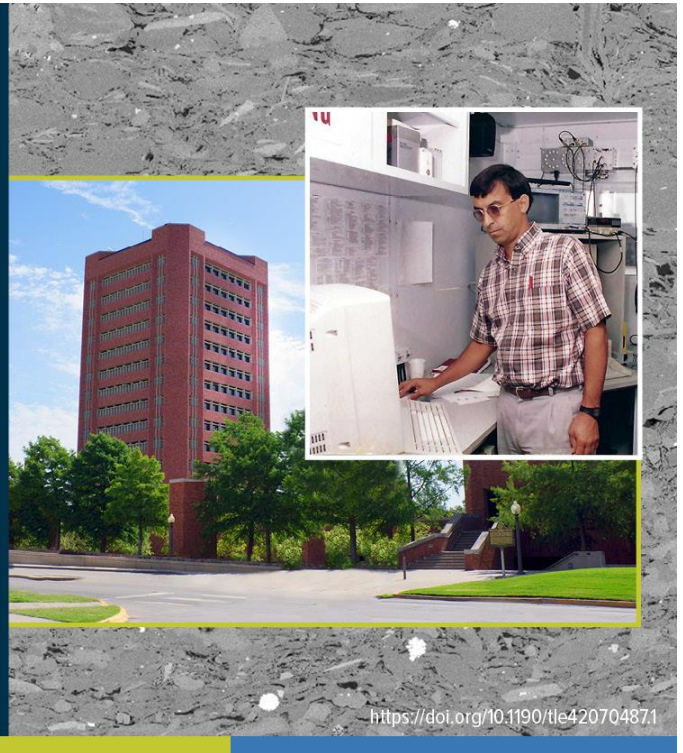
WORKSHOP

FROM MEASUREMENTS TO THEORY:

*Adventures in Rock Physics,
Petrophysics, and Engineering*

Honoring the Contributions of Professor Carl H. Sondergeld

27–29 Mar '24 • Norman, OK, USA



TECHNICAL PROGRAM WITH ABSTRACTS



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SEG/SPE/SPWLA Workshop:
From Measurement to Theory: Adventures in Rock Physics, Petrophysics, and Engineering
Honoring the Contributions of Professor Carl H. Sondergeld

Technical Program

Wednesday 27 March 2024		
Time	Presentation	Presenters*
7:30–8:30	Registration and Breakfast Buffet	
8:30–8:45	Welcome and Introduction	<i>Co-Chair: Ronny Hofmann, Shell</i>
8:45–9:30	<i>Carl's Life</i>	<i>Chandra Rai, Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma</i>
9:30–12:15	Session 1: Petrophysical Characterization <i>Session Chairs: Joe Comisky, Nick Drenzek</i>	
9:30–9:55	Organic Matter, Porosity, and Gas Production in the Marcellus Shale	<i>L. Taras Bryndzia*¹, R. Hofmann² (1. Independent Geoscientist, 2. SIEP Inc.)</i>
9:55–10:20	Review of Gas Injection EOR in Unconventional Reservoirs: From the Optimization of Injection-Production to the Understanding of Main Drive Mechanism	<i>Son Dang*¹, C. Rai¹, C. Sondergeld¹ (Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma)</i>
10:20–10:45	Enhancing Oil Recovery from Fractured Reservoirs with a Wettability Modifier	<i>Luis E. Zerpa*¹, A. Siddique¹ (1. Colorado School of Mines)</i>
10:45–11:00	BREAK	
11:00–11:25	<u>Microseismic</u> Evolution, Fault Reactivation, and Stress Heterogeneity of Crustal Faults at Pawnee, Oklahoma	<i>Segun S. Bodunde*¹, J. Jiang¹ (1. School of Geosciences, The University of Oklahoma)</i>
11:25–11:50	New Dynamic Measurements for Permeability from the Reservoir to Core Scale	<i>Mark Proett*¹ (1. Mark Proett Consulting LLC)</i>
11:50–12:15	Multi-Scale Characterization for Unconventional Reservoir Interpretation: Realizing Improved Geophysical and <u>Geomechanical</u> Integration	<i>Mark G. Kittridge*¹, S. Plitzuweit¹, J. Estrada¹, S. Olgati¹, J. Kessler¹ (1. Occidental Petroleum Corporation)</i>
12:30–13:30	Lunch Sponsored by <u>NuTech</u> Energy	

13:30-16:00	Session 2: Rock Physics <i>Session Chairs: Tad Smith, Mark Kittridge</i>	
13:30-13:55	Estimating Elastic Moduli of Solid Organic Matter in Unconventional Reservoirs	<i>Paritosh Bhatnagar*¹, L. Vernik² (1. Ikon Science, 2. University of Houston)</i>
13:55-14:20	New Method of XRF/XRD Based Mineralogy for Estimating Matrix-Bulk-Modulus, Biot's-Coefficient, and Effective-Stress Using VRH and HS Bounds Models of Rock Physics and Petrophysics	<i>Umesh Prasad*¹, N. Alarcon¹ (1. Baker Hughes)</i>
14:20-14:45	Rock Physics AVO Inversion Using Gauss-Newton Optimization with Analytical Derivatives	<i>Jorlivan Correa*¹, D. Grana² (1. Petrobras, 2. University of Wyoming)</i>
14:45-15:10	Integration of Rock Physics Models in Seismic Petrophysical Inversion Using Bayesian Physics Informed Neural Networks	<i>Peng Li*¹, D. Grana¹, M. Liu¹ (1. University of Wyoming)</i>
15:10-15:35	Machine Learning Application for Joint Rock Physics Model Optimization, Facies Classification, and Compaction Modeling	<i>Gabriela D'Aubeterre*¹, R. Filograsso¹, A. Snowling¹ (1. Ikon Science)</i>
15:35-16:00	Multi-Ensemble Bayesian Analysis to Determine the Rock-Physics Model with the Highest Probabilistic Fit	<i>Kyle T. Spikes*¹, M. Sen¹ (1. The University of Texas at Austin)</i>
16:00-16:30	BREAK	
16:30-18:00	Session 3: Poster Session <i>Session Chairs: Mark Curtis, Ronny Hofmann</i>	<i>3 rounds of 20-minute presentations with 2-minute introductions before each round</i>
	Combined X-Ray and Electrical Resistivity Tomography to Quantify Capillary-Pressure Effects During Mud-Filtrate Invasion	<i>Mohamad Abdo*¹, C. Torres-Verdin¹ (1. The University of Texas at Austin)</i>
	Effective Elastic Properties of Composite Granular Rocks Using Particulate Discrete Element Modelling	<i>Prasad Pothana*¹, F. Garcia², K. Ling¹ (1. Department of Energy and Petroleum Engineering, University of North Dakota, 2. Civil and Environmental Engineering, University of Michigan)</i>
	Advancing Reservoir Characterization in Clay-Rich, Thin Bedded Clastic Reservoirs through the Extended Thomas-Stieber Model	<i>Tariq Saihood*¹, M. Myers¹, L. Hathon¹ (1. Rock Mechanics, Petrophysics, and Sample Characterization Laboratory, University of Houston)</i>
	Vertical Permeability Estimation in Multilayer Anisotropic Reservoirs in the Natih Formation, Oman	<i>Adeyemi A. Adetosoye*¹, C. Johnson¹ (1. Occidental Petroleum Corporation)</i>
	Natural Fractures in the Niobrara Formation, Redtail-East Pony Field Areas, Denver Basin, Colorado	<i>Stephen A. Sonnenberg*¹, C. Matson¹ (1. Colorado School of Mines)</i>
	Early EUR Indicator for Permian Basin Unconventional Resources Based on Hourly Flowback Data	<i>Xueying Xie*¹, S. Liu¹, C. Leiker¹ (1. Occidental Petroleum Corporation)</i>
	Hydromechanical Impacts of Rock Layer Heterogeneity on Poroelastic Deformation and Fluid Flow in Geothermal Fields: A 3D Finite-Element Modeling Approach	<i>Ganiyat Shodunke*¹, J. Jiang¹, S. Bodunde¹ (1. Department of Geosciences, The University of Oklahoma)</i>
	Seal-Leakage Analysis and Forward Modeling of a CO ₂ Sequestration Site	<i>Maureen James*¹, M. Prasad¹ (1. Department of Geophysics, Colorado School of Mines)</i>
	Multiparameter Statistical Rock-Physics Simulations for Igneous Rocks Intercalated with Subsalt Carbonate Reservoirs	<i>Vitor Mello*¹, K. Spikes¹ (1. Jackson School of Geosciences, The University of Texas)</i>
18:00-19:00	Networking Reception: Thurman J. White Forum Building	

Thursday 28 March 2024		
Time	Presentation	Presenters*
7:30–8:20	Breakfast Buffet	
8:20–8:30	Welcome	<i>Co-Chair: Joe Comisky, Devon Energy</i>
8:30-12:30	Session 4: Characterization and Image Based Petrophysics <i>Session Chairs: Chandra Rai, Kent Newsham</i>	
8:30-8:55	A Novel Technique for Determination of Formation Water Salinity Through Ionic Exchange	<i>Abdelali Guezei*¹, S. Mamoudou¹, S. Dang¹, C. Rai¹ (1. Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma)</i>
8:55-9:20	Integration of Mercury Injection Capillary Pressure Derived Properties for Improved Prediction of Reservoir Performance	<i>James Greene*¹, K. Newsham¹, M. Pavlovic¹ (1. NuTech Energy)</i>
9:20-9:45	Rapid Characterization of Quantitative Mineralogy Using FTIR Analysis	<i>Rieko Adriaens*¹, G. Mertens¹, B. Paredis¹ (1. Qmineral Analysis & Consulting)</i>
9:45-10:10	Characterizing Variable Pore Systems in a Heterogeneous Carbonate Reservoir (Lower Habshan) in UAE: Using Multiple Data Collection Approaches	<i>Sweta Bose*¹, J. Moreno¹ (1. Occidental Petroleum Corporation)</i>
10:10-10:30	BREAK	
10:30-10:55	Small Things with Huge Impact: 15 Years of Learnings from Imaging Shale Microstructure	<i>Mark E. Curtis*¹ (1. The University of Oklahoma)</i>
10:55-11:20	Elastic Properties Derived Directly from SEM Images in Unconventional Plays	<i>Deepak Devegowda*¹, H. Yang¹, M. Curtis¹, C. Rai¹, C. Sondergeld¹ (1. Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma)</i>
11:20-11:45	Estimating the Impact of Clay Conductivity on the Electrical Properties of Shaly Sands Using Digital Rock Physics	<i>Bochao Zhao*¹, R. Hofmann¹, J. Freeman¹ (1. Shell International E&P)</i>
11:45-12:10	Petrophysical Properties from Images	<i>Ronny Hofmann*¹, N. Saxena¹, J. Freeman¹ (1. Shell International E&P)</i>
12:30-13:30	Lunch Sponsored by Ikon Science and Devon Energy	
13:30-16:00	Session 5: NMR and Measurements <i>Session Chairs: Nick Drenzek, Mark Proett</i>	
13:30-13:55	NMR Interpretation in Unconventional Reservoirs: Application to QC/Guide for Log Calculated Total Water Saturations in the Texas Delaware Basin	<i>Jing Li*¹, M. Pavlovic¹ (1. Occidental Petroleum Corporation)</i>
13:55-14:20	Overview of NMR/MRI of Rocks	<i>Michael Dick*¹, T. Kenney¹, D. Veselinovic¹, D. Green¹ (1. Green Imaging Technologies)</i>
14:20-14:45	Practical Applications for NMR Core Analysis	<i>Michael Dick*¹, T. Kenney¹, D. Veselinovic¹, D. Green¹ (1. Green Imaging Technologies)</i>
14:45-15:10	Understanding Non-Exponential Characteristics of NMR Responses from Unconventional Source Rocks and Impacts to Core Analysis	<i>Z. Harry Xie*¹ (1. Core Laboratories)</i>
15:10-15:35	Investigating Rock Anisotropy Effects on Acoustic and Electrical Properties through Angle-Dependent Measurements	<i>Domenico Crisafulli*¹, D. Olszowska¹, C. Torres-Verdin¹ (1. The University of Texas at Austin)</i>

15:35-16:00	Comparative Experimental Study of Ultrasonic Attenuation Measurements in H ₂ and CO ₂ -Saturated Fontainebleau Sandstones	<i>Gabriel C. Unomah^{*1}, M. Myers¹, D. Myers², L. Hathon¹, J. Adolrazzag³, R. Stewart³ (1. Department of Petroleum Engineering, University of Houston, 2. Metarock Laboratories, 3. Department of Earth and Atmospheric Sciences, University of Houston)</i>
16:00-16:30	Break and Walk to Bizzell Museum	
16:30-17:30	Tour: History of Science Collections, Bizzell Memorial Library, The University of Oklahoma	
17:30-19:00	Networking Reception: Thurman J. White Forum Building	

Friday 29 March 2024		
Time	Presentation	Presenters*
7:30-8:20	Breakfast Buffet	
8:20-8:30	Welcome	<i>Co-Chair: Mark Proett, Petroleum Engineering Consultant</i>
8:30-12:30	Session 6: Sustainable Energy Developments <i>Session Chairs: Kent Newsham, Mark Curtis</i>	
8:30-8:55	Methane Gas Hydrate Precipitation Modeling Using Digital Rock Physics	<i>Daniel Rathmaier^{*1}, L. Zerpa¹ (1. Colorado School of Mines)</i>
8:55-9:20	Evaluation of Hydrogen Interaction with Mudrocks: Implication for Geo-Storage	<i>Sidi Mamoudou^{*1}, S. Dang¹, C. Rai¹ (1. Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma)</i>
9:20-9:45	Characterizing the Geomechanical and Geochemical Suitability of Saline Aquifers for Hydrogen Storage	<i>Mohamed Sarhan^{*1}, L. Hathon¹, M. Myers¹ (1. University of Houston)</i>
9:45-10:00	BREAK	
10:00-10:25	Analysis of Seismic Anisotropy in Kevin Dome, Montana: A CCUS Project — From Core to Seismic	<i>Alexandro Vera-Arroyo^{*1}, H. Bedle¹ (1. The University of Oklahoma)</i>
10:25-10:50	Rock Response to CO ₂ Fracturing and its Implication for Carbon Sequestration	<i>Blessed C.A. Amoah^{*1}, C. Rai¹, M. Curtis¹, S. Dang¹ (1. Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma)</i>
10:50-11:10	The Effect of Anisotropy on scCO ₂ Transport in Shales: Implications to Carbon Geostorage	<i>Felipe Cruz^{*1}, S. Dang¹, C. Rai¹ (1. Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma)</i>
11:15-12:00	Closing Remarks	
12:00-12:30	Lunch	
12:30-14:00	Tour (Optional): Integrated Core Characterization Center, petrophysics laboratory, Petroleum School	<i>Chandra Rai, Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma</i>



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Organic Matter, Porosity and Gas Production in the Marcellus Shale

L. Taras Bryndzia¹ & Ronny Hofmann²

1 - Independent Geoscientist, Houston, Texas

2 – Research Petrophysicist, SIEP Inc., Houston, Texas

Abstract

A common paradigm of unconventional shale gas resource development posits that the highest TOC source rock facies most likely represent the elusive “sweet spot” of high gas saturation and porosity. This is based on the assumption that the highest TOC intervals (often characterized by the lowest bulk density and highest Uranium intervals), also host well-developed shale porosity, the holy grail of storage capacity in low permeability shale source rock reservoirs.

The TOC ‘sweet-spot’ paradigm for shale gas was tested in two Marcellus field trials in PA, based on the novel deployment of a Down Hole Raman Reservoir Spectrometer (DRRS) system. The technical objective of both field trials was to identify and quantify hydrocarbon resource density, as well as the most favorable location(s) for the placement of horizontal development wells.

The DRRS system measured, in real-time, the concentrations of both free and dissolved methane gas in a water-filled well bore. The results of the field trials showed that of the three organic-rich facies in the Marcellus gas shale (Basal, Middle, and Upper zones), high TOC was not the controlling element for highest methane resource density or rate of gas production.

Various mechanisms that might also impact gas storage and production, such as compaction of OM pores by overburden stress are also discussed. However, the primary control on rate of gas rate production was the presence of a well-developed OM porosity network that delivered significant volumes of free gas to the well bore, even in the absence of hydraulic fracture stimulation.

Based on measured plume concentrations observed in our field trials, the estimated resource density i.e., the mass of methane per volume of rock, is highest in the Upper TOC-rich Marcellus shale. Based on our DRRS results we would prioritize completion of the Upper zone with a horizontal producer.

Review of gas injection EOR in unconventional reservoirs: from the optimization of injection-production to the understanding of main drive mechanism

Son Dang, Chandra Rai, Carl Sondergeld

Mewbourne School of Petroleum and Geological Engineering
University of Oklahoma

The primary recovery of hydrocarbons (HCs) in organic rich shales is relatively low, ranging from 5-10%. In light of the rapid decline of production, application of EOR techniques becomes an attractive approach. Considering the low matrix permeability and reservoir containment, cyclic gas injection EOR, huff-n-puff, is used by the industry as an optimal method to further extract HCs from shales.

In this comprehensive study, we used three experimental approaches: 1) measured Minimum Miscibility Pressure, oil swelling and bulk oil-gas diffusion using Vanished Interfacial Tension technique coupled with a high pressure glass window cell, 2) used passive Nuclear Magnetic Resonance (NMR) technique to estimate oil and water production from huff-n-puff experiments on crushed samples, while changing different parameters, such as injection pressure, exposure time, number of cycles, and injectate composition, 3) used the real time NMR monitoring technique to observe the dynamic interaction between the injectate and HCs in the rock during huff-n-puff experiments on plug samples for evaluation of the relative contribution of different drive mechanisms.

Theoretical studies suggest that within nanoporous media, diffusion can be the dominant mass transport, which is governed by bulk fluid diffusivity and matrix tortuosity. We observed incremental recovery during the injection phase, due to oil swelling and gas vaporization, which is governed by diffusion. Moreover, the incremental recovery during the production steps follows the same projected trend observed through the injection steps, which suggests advective flow has minimal contribution to oil recovery. For a specific sample, when the confining stress was applied, the recovery reduced from 30% to 5%, which highlighted the importance of the open surface area for the diffusion mechanism.

We recommend matrix tortuosity, fluid diffusivity, and fractured surface area should be considered to optimize of HCs output in tight reservoirs, including both primary production and EOR.

Enhancing Oil Recovery from Fractured Reservoirs with a Wettability Modifier

Afrah K. Siddique, Luis E. Zerpa, Colorado School of Mines

Petroleum Engineering Department, Colorado School of Mines, 1500 Illinois St., Golden, Colorado, USA

Enhanced Oil Recovery methods using injection of water-soluble chemical additives (e.g., surfactants), has potential for increasing oil recovery from fractured reservoirs. The main factors that affect oil recovery from fractured reservoirs are the rock wettability, reservoir heterogeneity, capillary trapping forces, and preferential flow of injected fluid through fractures. This paper presents the evaluation of a wettability modifying agent, a mutual solvent in water and oil phases, that when injected in fractured reservoirs, promotes water imbibition into the rock matrix while enhancing oil recovery. The injection of a wettability modifying agent, reduces the interfacial tension between oil and water, alters the rock wettability, and dissolves in both oil and water to enhance oil recovery. This study focuses on the application of a mutual solvent injection in fractured cores. Preliminary evaluations included measurements of interfacial tension and rock wettability using a tensiometer. Coreflooding experiments were conducted to measure the incremental oil recovery by the dissolution agent injection in an intact (unfractured) core sample and in a fractured core sample. In this study we used 3-Pentanone as the dissolution agent, an efficient wettability modifier and a mutual solvent for oleic and water phases. The effects of changing concentration of 3-Pentanone and including a soaking period before continuing injection were also investigated. The experiments were performed on core samples of the Colton sandstone formation from Central Utah. The incremental oil recovery by using the dissolution agent in low salinity brine from unfractured and fractured core was 7% and 22%, respectively. The incremental oil recovery using 3-Pentanone can be explained by the combination of various mechanisms that includes a slight reduction in interfacial tension, alteration of rock wettability from water-wet to strongly water-wet, and partitioning of ketones into the oil phase that reduces the oil viscosity and increases oil mobility.

New Dynamic Measurements for Permeability from the Reservoir to Core Scale

Mark A. Proett, Petroleum Engineering Consultant

Formation rock permeability and its anisotropy is one of the essential properties impacting reservoir development. It is an inherently dynamic property, and the most accurate measurements employ flow transients within the rock's pore space. Estimates can be made using static measurements such as porosity and density, but these methods usually rely on a correlation with dynamic measurements.

In this talk some of the new dynamic methods are presented for different dimensional scales. Reservoir scale and even interval scale measurements are typically performed during the drilling, completion, and even production stages. The technology used can have different effective scales, such as a wireline formation pretest, formation interval transient testing, and full-scale reservoir tests. The formation wireline pretest is more to the scale of a core test and has some similarities. The rock near the probe is contaminated with drilling fluids that impact the measurement much like a core that must be cleaned and properly prepared to make a most accurate measurement. Even interval and reservoir testing can be impacted by the near well bore drilling damage. These issues are being addressed by some new and innovative measurement techniques that will be reviewed for discussion.

**Multi-scale characterization for unconventional reservoir interpretation:
Realizing improved geophysical and geomechanical integration**

Mark G. Kittridge, Shelby Plitzuweit, Jorge Estrada, Silvia Olgiati, & James Kessler

The use of bulk density and elastic (P- and S-wave velocities) properties are essential to create a robust seismic-based (3D) reservoir characterization methodology, as well as establishing mechanical properties for completion design in unconventional and tight reservoirs. The study developed sub-regional (field-scale) models using available vertical pilot well data, where additional high-end formation evaluation measurements were often available. We create a suite of models connecting available core measurements and key formation variables (clay, TOC) to density and velocity, and thus derived properties (E , ν). Our workflow applies both data-driven and model-based elements, recognizing the primary mineral components (quartz, calcite) and the combined impact of clay and TOC on key geophysical variables. The consistent use of a robust mineral-based formation evaluation model provides a natural platform for multi-well integration and characterization of essential geophysical properties. Our bench-specific elastic properties models enable robust QC and forward modeling of dynamic variables useful for both seismic inversion and as inputs to stress and unconventional reservoir completion design. In this paper we share results from a unique pilot study where we integrate multi-scale measurements from core- and well-measured properties in two distinct development benches. High vertical resolution lab data agree very well with novel measurements made with a new LWD multi-physics acoustic platform and reveal an important degree of (vertical) heterogeneity missing from conventional well log measurements. We measured directional dynamic and static core properties using a lab protocol that enables polar (TI) anisotropy characterization and find a wide range of behavior between the two geologic benches. A portable modeling workflow was designed that is easily adapted to new core data and ensures we account for anisotropy in our geophysical characterization and completion design. Deliberate multi-scale interrogation provides a framework for realizing real integration and understanding of geophysical and geomechanical properties in unconventional reservoirs.

New method of XRF/XRD based mineralogy for estimating matrix-bulk-modulus, Biot's-coefficient and effective-stress using VRH and HS bounds models of rock physics and petrophysics.

By: Umesh Prasad and Nora P. Alarcon, Baker Hughes

Abstract:

In any civil, mining, or petroleum application the concept of Stress (or strain) under the influence of load or displacement is the foundation of geomechanical investigations. However, since the subsurface geological formation consists of pore, voids and fluid pressure, it is the 'effective stress' and not the 'total stress' which controls the deformation, fracture, failure, and production decline characteristics including dynamic acoustic wave propagation, porosity and permeability in rock formations. Effective stress is also useful in any triaxial core testing in the laboratory for calibrating log-based estimations. Effective stress (σ') is expressed as difference between total stress (σ) applied and existing pore pressure (P_p) multiplied by Biot's coefficient, α . Traditionally, ' α ' is expressed in terms of bulk modulus (K_b) and matrix bulk modulus (K_{ma}) as in $1 - K_b/K_{ma}$. The K_b is estimated by acoustic logs as well as measured by triaxial test in the laboratory. However, K_{ma} is difficult to measure, especially in low porosity or low permeable formation.

Current advancement in Surface-logging at the rig-site has enabled use of portable X-ray fluorescence (XRF) and X-Ray diffraction (XRD) tools which are low cost and quick turn-around time. These tools and their analysis have been used in geomechanical analysis first time. The XRF/XRD analysis obtained from different wells were analyzed. The mineralogy obtained from XRF are compared with that obtained from XRD and laboratory measurements. The Rock-physics principles of upper-bound Voigt (1910), lower-bound Reuss (1929), and an average of two, Hill (1963) as well as upper and lower Hashin-Shtrikman (1963) bounds were compared. Finally, the Biot's coefficient obtained from above models were compared with porosity-based estimations and laboratory testing. The effective stress obtained from above gives a great confidence in modeling and simulation of sub-surface integrity evaluation, calibrating horizontal stresses for both geothermal, unconventional, and hydraulic fracture design applications.

Rock physics AVO inversion using Gauss-Newton optimization with analytical derivatives

Jorlivan Correa¹ and Dario Grana²

¹ Petrobras; ² University of Wyoming

Seismic petrophysical inversion aims to predict petrophysical properties, such as porosity, lithology, and fluid saturation, given a set of seismic measurements by combining inverse methods with AVO and rock physics models. Generally, the inversion process is performed in two steps: first, seismic data are inverted for elastic properties, then petrophysical properties are predicted from elastic attributes. Alternatively, the inversion can be performed in a single loop, by defining a geophysical model that maps the petrophysical variables into the seismic response based on a linearized approximation of Zoeppritz equations and a rock physics model. The inversion can then be performed using deterministic or stochastic methods. The proposed geophysical model allows analytically computing the Jacobian matrix of the objective function of the inversion. Indeed, with this geophysical model, the partial derivatives of the seismic properties with respect to the petrophysical properties can be expressed in a closed form. The choice of the rock physics relations depends on the geological characteristics of the reservoir rocks. In our implementation, we tested the inversion with the following model: the seismic forward model is a convolutional operator of a known wavelet and Aki-Richards linearized approximation, whereas the rock physics model is Dvorkin's soft sand model based on Voigt-Reuss-Hill average for the rock matrix properties, Reuss average for the fluid mixture properties, Hertz-Mindlin model and modified Hashin-Shtrikman bounds for the dry-rock properties, and Gassmann's equations for the saturated-rock properties. However, the proposed inversion method can be extended to any rock physics model and any AVO formulation. Based on this model, we develop a Gauss-newton inversion using Levenberg-Marquardt method, a nonlinear optimization algorithm that estimates the Hessian matrix using the sum of outer products of the gradients. The proposed method has been tested and validated on a synthetic dataset simulated based on real data measured offshore Norway.

Integration of rock physics models in seismic petrophysical inversion using Bayesian physics informed neural networks

Peng Li, Dario Grana, and Mingliang Liu

ABSTRACT

The inversion of seismic data for petrophysical properties requires the integration of a rock physics model in the inverse method. One of the main challenges in using rock physics models in seismic inversion is that the model parameters, such as coordination number or aspect ratio, are often uncertain. In this work, we develop an inverse method for seismic petrophysical inversion based on deep learning algorithms, where the rock physics model parameters are learned by the network from a training dataset. We adopt a deep learning approach that integrates physical models, with known or unknown parameters, in a neural network algorithm, namely a physics-informed neural network (PINN), where the objective function of the inversion is defined by the Kullback–Leibler divergence and it includes both data-dependent and physics-dependent loss terms. Because our goal is not only to predict the most likely model but also to quantify the uncertainty of the model predictions, we implement the PINN in a stochastic deep learning framework. The method combines a Bayesian approach with a PINN algorithm via variational inference, namely Bayesian physics-informed neural networks via variational inference (BPINN-VI), and it aims to estimate probability distribution of the training parameters of the neural network by proposing a family of probability density functions and finding the candidates that are close to the target distribution. The inversion is applied to a seismic inversion problem for the estimation of petrophysical properties from seismic data, where we assume that the rock physics relations between petrophysical and elastic properties are known but the parameters of the models are unknown. The outcomes of the inversion include the most likely model of petrophysical properties and their uncertainty as well as the probability distribution of the rock physics model parameters.

Machine learning application for joint rock physics model optimization, facies classification and compaction modeling

Rosalena Filograsso¹, Andrew Snowling¹, Gabriela D'Aubeterre¹. 1 Ikon Science.

Abstract:

Petrophysical facies interpretation and Rock Physics modelling are two fundamental stages in any quantitative interpretation workflow. To perform these stages accurately and efficiently, data exchange and cooperation with different subject matter experts are critical. As collaboration tends to increase complexity, reliable automation is a needed tool for fast and smooth delivery. Automation for petrophysical facies interpretation translates to machine learning algorithms that, at the current state-of-the-art, are not able to fully capture compaction trends. The error added to such automated ML interpretation may propagate in the Rock Physics modelling and ultimately in the inversion.

To overcome these current ML limits, we have developed the Rock Physics Machine Learning (RPML) toolkit that takes compaction trends into account in the inversion for petrophysical facies and rock physics model parameters. The toolkit outputs the interpreted petrophysical facies logs, the petro-elastic depth trends and the rock physics parameters as optimized by the inversion. The RPML toolkit allows saving of both the prior, regionally bound models, as well as the posterior calibrated rock physics models, thus optimizing the data exchange among teams. In fact, a regional prior model can be tested, approved, and applied to different areas in the same region, assuring consistency and accuracy across different projects. The case study presented here is set in the Central North Sea, within the Forties Field and illustrates the flexibility of the application, consistency with manual facies interpretation, and value of cross-disciplinary integration.

Combined X-Ray and Electrical Resistivity Tomography to Quantify Capillary-Pressure Effects During Mud-Filtrate Invasion

Mohamad Abdo and Carlos Torres-Verdín, The University of Texas at Austin

ABSTRACT

Overbalanced drilling gives rise to mud filtration, which is the separation of the solid and liquid phases present in drilling mud. Mud filtrate invades permeable formations at an ever-slowing flow rate. In addition, the velocity of the invasion front decreases due to the cylindrical nature of the flow and its increasing geometric factor. According to fractional flow theory, slower invasion fronts exhibit more pronounced capillary smoothing. Understanding and quantifying capillary-smoothing effects helps in the interpretation of fluid-sampling and formation-tester measurements. Rapid micro-Computed Tomography (μ CT) can visualize the extent of the invasion in heterogeneous formations without fully sensing capillary-smoothing effects. This study implements the use of Electrical Resistivity Tomography (ERT) using four thin aluminum ring probes to quantify capillary smoothing of the front of mud filtrate invading cylindrical samples subject to X-ray μ CT scanning. ERT has been traditionally used in the geotechnical industry to quantify soil moisture and aquifer salinity. Corresponding conversion equations have been established in the industry. However, due to the uniqueness of the experimental system and the time-dependent electrical shape factor, our study replicates experimental results numerically via the Finite Element Method (FEM). Matching experimental measurements with numerical simulations enables the estimation of electrical resistivity variations across the sample. Sensitivity analysis on the results obtained shows that after 10 minutes of brine injection into a cylindrical borehole of a cylindrical sample, voltage readings drop by 1 Volt as the capillary entry pressure increases from 1 psi to 10 psi. These results, in combination with Leverett's J function, enable the differentiation between a high-permeability formation with low capillary-smoothing effects and a low-permeability formation with high capillary-smoothing effects.

Title: Effective Elastic Properties of Composite Granular Rocks Using Particulate Discrete Element Modelling

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Abstract

The mechanical behavior of composite granular rocks is a multifaceted phenomenon relevant to various geoengineering applications. Traditional homogenization models and continuum mechanics-based numerical methods often fall short of accurately capturing the intricacies of granular materials. Granular materials exhibit heterogeneity and arching mechanisms governing the force networks that ensure system stability. Unlike continuum-based approaches, particulate discrete element methods have an advantage in assessing effective material properties by considering material heterogeneity and grain-level physical interactions. This study evaluates effective elastic properties using particle flow code using flat-joint contact law for composite binary mixtures with a stiff inclusion embedded within a matrix material. We created digital models of granular rocks to simulate the properties of a moderate-strength sandstone matrix. Stiff inclusions were incorporated in two geometric configurations: i) as structural inclusions and ii) as laminated inclusions. Our numerical experiments involved subjecting the models to unconfined compressive, isotropic, and shear stress loading to analyze the material's anisotropic elastic properties. This included the estimation of Young's modulus, Poisson's ratio, bulk modulus, and shear modulus in all directions.

Our findings closely adhere to the Voigt-Reuss and Hashin-Strikman bounds within their specific conditions, demonstrating the promising application of the discrete element method in the analysis of composite materials. Additionally, the results adhere to the principles of generalized Hooke's law under small strain conditions. A detailed analysis is carried out on the stress-strain distribution within the specimen, examining its deviation from the idealized concepts such as iso-stress and iso-strain behavior. Additionally, our research provides an in-depth analysis of the distinctive stress-evolution and damage-evolution characteristics exhibited by different geometrical configurations of inclusions. These results offer invaluable insights into the mechanical behavior of composite granular rocks and underscore the potential applications of the discrete element method in addressing rock physics modeling problems encountered in petroleum engineering.

Advancing Reservoir Characterization in Clay-Rich, Thin Bedded Clastic Reservoirs through the Extended Thomas-Stieber Model

Tariq Sahood¹, Michael T. Myers¹ and Lori Hathon¹

¹Rock Mechanics, Petrophysics and Sample Characterization Laboratory, University of Houston

The extended Thomas-Stieber method has emerged as a pivotal tool in the evaluation of clay-rich, thin bedded clastic reservoirs, addressing the inherent limitations of conventional models by incorporating sonic log data as a crucial third dimension. This extended model, reliant on three fundamental variables—volume fractions of dispersed clay (X_d), laminated clay (X_l), and structural clay (X_s)—initiates with iterative derivations of equations for total porosity, gamma ray response, and effective P-wave slowness. The formulation integrates heterogeneous clays and quartz into the matrix, employing Wyllie's time-average equation and the Thomas-Stieber model.

A pivotal aspect of this model is the utilization of a three-dimensional grid template, where gamma ray, bulk density, and sonic log responses serve as axes, facilitating the visualization of total porosity evolution concerning diverse clay/shale components. Noteworthy observations include distinct correlations between clay types and total porosity, with structural clays displaying a positive association and laminated clays exhibiting an inverse relationship. Dispersed clays manifest inverse associations with both porosity and P-wave slowness. The integration of acoustic well log data enhances sensitivity to structural clays.

The calibrated extended Thomas-Stieber model is applied to well log data from a shaly sand reservoir, yielding insights into volumetric fractions of dispersed clay (X_d), structural clay (X_s), quartz (X_q), and laminated clay (X_l). The integration of sonic log responses offers a nuanced assessment of clay distributions and their influence on porosity in shaly sands. This refined model contributes to heightened accuracy in reservoir characterization and simulation, thereby advancing reservoir management and optimizing hydrocarbon recovery strategies. Consequently, this study presents a novel extension of the Thomas-Stieber model, integrating sonic log responses for a comprehensive evaluation of clay distributions and their intricate impacts on porosity in thin bedded clastic reservoirs.

Vertical Permeability Estimation in Multilayer Anisotropic Reservoirs in the Natih Formation, Oman

A.A. Adetosoye, & C.N. Johnson

Accurate input of reservoir permeability and water saturations in the numerical simulation models are of paramount importance for production performance evaluation. For an oil reservoir under waterflooding, the horizontal and the vertical permeabilities have a big influence on the areal and vertical sweep efficiencies. In addition to correcting S_w due to resistivity anisotropy effects, a new workflow is proposed to improve the estimation of the vertical permeability in the Natih formation, which is a multilayered and heterogenous reservoir found in Oman, through resistivity anisotropy modeling. In this work, the vertical permeability anisotropy (K_v/K_h) was estimated from the cores taken from a vertical pilot hole and correlated with the resistivity anisotropy ratio (R_h/R_v) measured from the horizontal (R_h) and vertical (R_v) resistivities. Using the R_h/R_v ratio, the vertical permeability from a lateral well with LWD logs was estimated at every foot interval. The R_h/R_v ratio was used to estimate the corrected resistivities, by modeling the phase attenuation measurements, using the vertical permeability anisotropy. We compare the resistivity ratios to those derived from the earlier technique of Adetosoye, A.A., Moreno, J.J., Al Ghuzaily, M.A., et al., 2023., to assess inversion techniques used, resulting water saturations and inferred impact on a numerical simulation model. The estimation of accurate S_w and vertical permeability as an independent variable based on resistivity anisotropy as proposed in this workflow can aid Reservoir Engineers in the assessment of the flow dynamics in heterogeneous anisotropic reservoir formations. Furthermore, it can help in accelerating the history matching process of large fields by improving the definition and degree of communication between layers. Additionally, this workflow can be used to estimate K_v/K_h in fields where core data is not available by correlating resistivity anisotropy with the nearby analogue fields having core data.

Natural Fractures in the Niobrara Formation, Redtail-East Pony field areas, Denver Basin, Colorado

Stephen A. Sonnenberg, Colorado School of Mines

Christopher Matson, Colorado School of Mines

The Upper Cretaceous Niobrara Formation is a chalk and marl deposit in the Denver Basin. The Niobrara is typically subdivided into chalk units (A, B1, B2, C, D, Fort Hays) separated by marl intervals. All members of the Niobrara are productive in the Redtail-East Pony fields. Cumulative production exceeds 60 million barrels of oil making the field one of the largest in the Denver Basin. The vertical depth to the Niobrara is 5,500 to 6,200 ft. Chalk beds have matrix porosities that range from 10-13% and permeabilities less than 0.1 md.

The chalk beds consist of coccolith-rich fecal pellets and pelagic forams with minor clay. Marls are similar in composition but have a larger clay and total organic carbon content. Larger inoceramid and oyster fossils are also present in both the chalks and marls. The marls have total organic carbon contents that range from 2 to 8 wt.%. The mechanical stratigraphy is quite different between chalk and marls. The chalks generally contain numerous vertical fractures whereas the marls behave in a more ductile fashion and generally lack visible vertical fractures.

Fractures present in the chalk beds are nearly vertical, mineral-lined closed to partially open. Fracture frequency is controlled by lithology. Chalks are most fractured marls less so. Mineral fill in fractures is calcite. Ninety percent of the fractures are within 10 degrees of vertical. Fracture length is a few inches to 10 inches in length. Fracture apertures are less than 50 microns. Fractures terminate at chalk-marl bed boundaries, stylolites, and ash layers. Natural vertical fractures in the Niobrara Formation in the study area are related to pore pressure and rock strength. These fractures are tension fractures and non-tectonic in origin. Similar fractures are present in the underlying Greenhorn Formation.

Wells in the Redtail-East Pony fields are horizontally drilled and hydraulically fracture stimulated. The hydraulic fracture stimulations may reopen the calcite filled fractures.

Early EUR Indicator for Permian Basin Unconventional Resources Based on Hourly Flowback Data

Xueying Xie, Shunhua Liu, Courtney Leiker, Oxy

During the initial flowback after fracturing unconventional wells, oil, gas, and water rates, as well as pressure data, are usually available on an hourly basis. Therefore, even though flowback may only take a few weeks, there are hundreds of available data points. We can utilize the data to predict early well performance for economics evaluation and further stretch to obtain an indication of long-term EUR. This paper outlines the innovative approach taken to predict short-term cumulative oil production and long-term oil EUR using hourly flowback data based on both physics and empirical correlations. Unconventional wells produce in the linear flow regime in early time. During this period, the production depends on pressure, fracture surface area, and reservoir and fluid properties, which could be lumped into one term and obtained from the linear slope of the RTA superposition time plot. Most wells exhibit a linear flow signature during the flowback period and thus the linear slope can be obtained. Hundreds of Permian wells with hourly flowback data were utilized to obtain the slope. The values of a combined term of the inverse of this slope and initial pressure for these wells were plotted against their corresponding one-year cumulative oil production and EUR. From this analysis, correlations for both one-year cumulative oil production and EUR have been established. Using these correlations based on historical wells, the one-year cumulative oil production and oil EUR of a new well can be estimated with just a few weeks of flowback data. This new method provides an early performance indication for new wells, new benches, or new areas. The application of the method could speed up completion trials and appraisal processes tremendously and optimize field development plans.

Hydromechanical Impacts of Rock Layer Heterogeneity on Poroelastic Deformation and Fluid Flow in Geothermal Fields: A 3D Finite-Element Modeling Approach

Ganiyat Shodunke, Junle Jiang, and Segun Bodunde

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Pore pressure changes due to the geomechanical interaction of reservoir rocks with circulating fluids can induce ground deformation with rates of mm to cm per year, especially during geothermal operations. Geophysical observations of deformation in major geothermal fields, e.g., North Brawley and Salton Sea, California, often rely on simplistic homogenous structural models to interpret subsurface pressure sources and fluid flow. However, rock layer heterogeneity (i.e., different rock physical properties for reservoir rocks and confining rock units) can control fluid movement and fluid-rock interactions, affecting subsidence and uplift in these areas. Here, we investigate the impacts of layered structural variations in geothermal fields in terms of porosity, permeability, confining rock thickness, and compaction on the total magnitude and temporal evolution of poroelastic deformation and pore pressure. We use a finite-element modeling approach and design 3D layered numerical models exploring these geologic scenarios under a constant production rate. To model deformation under compaction, we use Athy's law with a compaction coefficient of 0.31 km^{-1} . For the effect of confining rock thickness on surface deformation, we investigate scenarios including shallow vs deep confining rock unit configurations. The porosity and hydraulic conductivity of simulated geothermal systems range from 2% to 40% and 10^{-6} m/s to 10^{-10} m/s , respectively. Our results suggest the influence of layered heterogeneity on poroelastic deformation and pore pressure modeling is significant, with steady-state rates of deformation and pressure change differing by up to 40% compared to a homogeneous model scenario. We show that permeability has a greater influence than compaction-influenced porosity on the magnitude and time evolution of subsidence and pore pressure in a geothermal setting. Our results have broad implications in guiding our efforts to better combine high-resolution surface and subsurface observations and understand and predict the hydromechanical behavior of complex reservoir structures within geothermal fields.

Title: Multiparameter statistical rock-physics simulations for igneous rocks intercalated with subsalt carbonate reservoirs

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Abstract:

Because most of the hydrocarbon reserves are concentrated in sedimentary rocks, the rock-physics of igneous rocks received very little attention. However, in two cases the delineation of igneous bodies is necessary. In one case they can be reservoir rocks. In another they can be intercalated with sedimentary reservoir and/or source rocks, such as in subsalt carbonate fields, offshore Brazil. Typically, the mineral composition and pore space are the main factors controlling the seismic response of rocks. Igneous mineralogy can be very complex with multiple combinations of felsic and mafic mineral concentrations. The porosities are typically low but can exceed 30% and can be associated to different pore type geometries. Thus, reliance on the biased statistics derived from sparse wells often produces inaccurate predictions. Here, we propose to combine statistical simulations and rock-physics modeling to assess the elastic response of igneous rocks, mitigating sampling bias. The calibrated rock-physics models are used to simulate unsampled, yet plausible, geological features that are likely to be absent or underrepresented in the observed data. The simulated data accounts for varying porosities, pore geometries and mineral composition. Through this workflow we reproduced the seismic properties of different igneous rocks with no dependence of direct measurements on these rocks. The results are validated using lithologic descriptions as wells as lab measurements in rock samples, such as ultrasonic velocities, porosity, and mineral composition. In this approach we overcome the lack of data, providing tools to a robust risk assessment in reservoir property prediction using seismic-derived properties.

Integration of Mercury Injection Capillary Pressure Derived Properties for Improved Prediction of Reservoir Performance

J. Greene, K. Newsham, M. Pavlovic

Disconnects between petrophysical volumetrics and observed reservoir performance can often be attributed to variability of dynamic rock properties across geologic facies. The acquisition of statistically significant core measurements that are time and cost effective can be challenging when characterizing heterogeneous depositional environments. Advanced measurement procedures and interpretation of mercury injection capillary pressure (MICP) data can yield information related to textural distributions, saturation history, and poroelastic behavior. The combination of measurement constrained values within empirical and physical models provides derivations of relative permeability and compressibility from Darcy to nano Darcy permeabilities. Modelling of fractional flow and residual hydrocarbon volumes is aided through the upscaling and integration of this core-based measurement data via petrophysical framework and engineering equations of state (EOS). Case study examples from both unconventional (Delaware Basin, New Mexico) and conventional (Chaco Este Field, Bolivia) reservoirs will be shown, with comparisons to production and complimentary core measurements for benchmarking.

Characterizing Variable Pore Systems In A Heterogeneous Carbonate Reservoir (Lower Habshan) in UAE: Using Multiple Data Collection Approaches

Sweta Bose¹ and Javier Moreno¹

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ABSTRACT

Reservoir heterogeneity in carbonate rocks can be related to the geological processes during original deposition as well as during post depositional diagenesis. The present study is focused on the Lower Habshan formation of Early Cretaceous age over southeast Abu Dhabi (UAE). Multiple data sources (e.g., NMR, thin sections, SEM, capillary pressure data) indicate that Lower Habshan formation has a wide range of pore structures and permeabilities which result in the heterogeneities of the reservoir rock. The lithology is primarily oolitic limestone with some dolomitic facies interspersed throughout. In general, this formation was deposited in a prograding carbonate ramp shoal system. Dominant reservoir facies are oolitic grainstones, skeletal peloid grainstones and grainy packstones of moderate to poor reservoir quality. Microporosity due to micritization has been observed in several locations and depths; Microporosity seems to be an integral part of the pore system that possibly degrades the rock fabric/reservoir quality. Micropores, along with macro pores and vugs drive the variability in the pore system (types, connectivity, and texture). Connectivity of macro pores and open vugs play a critical role in the reservoir flow network. Our current workflow includes building petrophysical models using simple triple combo logs integrating responses from NMR, nuclear spectroscopy, and rock data RCA. This provides an overall characteristic of the reservoir in terms of lithology, porosity, and water saturation. However, understanding the inherent complexity of this carbonate reservoir proves challenging when relying solely on the general petrophysical model. The next phase focuses on integrating results from pressure data, advanced NMR, image logs, CT, thin sections, SEM, and capillary pressure data to evaluate the pore system, discern reservoir heterogeneity and relate to the recent well test results. Results and observations conclude that ingrained heterogeneity originating from the variable pore system has made the appraisal of Lower Habshan formation quite unique.

Estimating the Impact of Clay Conductivity on the Electrical Properties of Shaly Sands using Digital Rock Physics

Bochao Zhao, Ronny Hofmann, Justin Freeman, Shell International E&P

The difference between the classic Archie and Waxman-Smiths equations is the accounting for conductive mineral phases in rocks, such as clays and shales rock fragments. If these effects are not accounted for, water saturation cannot be accurately estimated in reservoirs with low reservoir water salinity and clays with high Cation Exchange Capacity (CEC).

The Waxman-Smiths model specifically incorporates the impact of clay conductivity and is one of the petrophysical models used to derive more accurate water saturations in shaly sands. The modified cementation exponent (m^*) and a measure of the excess conductivity (Q_v) are two parameters in the Waxman-Smiths equation to understand the effect of clay and salinity conductivity. These two parameters are usually determined in the lab by a so-called co-cw experiment, which involves measuring the bulk conductivity of a core plug saturated with brines with different salinities.

In the talk, we show that a 'digital' co-cw experiment can be performed using pore scale simulations. This digital twin of the co-cw experiment can significantly accelerate petrophysical data delivery by eliminating sample cleaning and equilibration time for different brine salinities. The micro-CT image is segmented into three discrete phases: pore, conductive and non-conductive minerals. Each of the phases is assigned an electrical conductivity and the bulk conductivity of the 3D volume is computed by solving Ohm's law at the pore scale. The properties of the conductive mineral phases are derived from a petrophysical understanding of the formation. The cementation exponent (m^*) and Q_v are calculated from the co-cw simulation, applying the same interpretation as for laboratory experiments. The presentation shows that Q_v is more sensitive to uncertainties in the clay properties compared to m^* . The difference between co-cw simulation and the lab experiments will also be discussed.

Title: NMR Interpretation in Unconventional Reservoirs: Application to QC/Guide for Log Calculated Total Water Saturations in the Texas Delaware Basin

Jing Li and Mitch Pavlovic
Occidental Petroleum Corporation

Nuclear Magnetic Resonance Logging (NMR) has proven to be an accurate and mineralogy-independent method for measuring total porosity and estimating permeability. OXY developed an in-house technique to estimate water saturation using NMR, which adopts a non-Archie approach and is independent of Archie m & n exponents and formation resistivity. NMR-based water saturation estimates have been evaluated, compared against core data, and implemented to guide/QC the Archie water saturation, which is particularly effective in the "low resistivity" siltstone/sandstone reservoirs in the Delaware Basin. While NMR is not necessary for every well, having NMR data available in the area can be helpful in guiding and quality-controlling the Archie water saturation.

Traditional petrophysical workflows estimate the water saturation using the Archie Equation, which relies on R_t (True Formation Resistivity) and electrical properties (m and n exponents). However, not only the presence of hydrocarbon but also the presence of kerogen and low-porosity rock could increase formation resistivity. Such elevated formation resistivity can result in a lower Archie water saturation, particularly in less favorable reservoir facies. Contrary, observed formation resistivity is typically very low in the clean, high porosity and permeability siltstone/ sandstone intervals across Permian Basin. In such cases, the Archie equation estimate of water saturation could be overly pessimistic ("low resistivity pay"), leading to less-than-optimal field development.

The NMR T_2 signal can be divided into eight porosity bins representing specific fluids. NmrBin1 and nmrBin2 are the smallest pores with the shortest T_2 time, corresponding to Clay Bound Water (CBW). As the T_2 distribution time increases, it indicates larger dominant pore sizes; Capillary Bound Fluids (BVI) (nmrBin3) and Free Water (nmrBin4) are likely to occupy these pores. NmrBin5-8 could be Hydrocarbons, which tend to occupy the most prominent pores in the rock, causing a shift in the T_2 distribution towards longer times. The volatile nature of hydrocarbons in unconventional reservoirs further enhances this shift due to diffusion-induced relaxation related to the presence of light oil (elevated GOR).

NMR estimated water saturation could refine the Archie water saturation, particularly in the "low resistivity" siltstone/sandstone reservoirs and less favorable reservoir intervals. It can improve the accuracy of Archie water saturation so that we have higher confidence in estimated oil-in-place. However, this technique may encounter issues when clean sands are water-filled and have low NMR SW readings, which leads to significant discrepancies with overly optimistic Archie water saturation values when m and n are low. The discrepancies combining formation resistivity can help interpret the fluids filled in the pore space, whether water or oil.

Overview of NMR/MRI of Rocks

Authors

M.J. Dick, T. Kenney, D. Veselinovic, and D. Green

Green Imaging Technologies, Fredericton, NB, Canada

Nuclear Magnetic Resonance (NMR) has long been used to examine rocks with either laboratory-based equipment or down hole measurements. NMR was first big innovation in the rock analysis was the discovery that the NMR relaxation properties directly correlate to the pore size of rocks. This further lead to NMR being used to quantify bound versus mobile fluid and led to relationships to calculate permeability. Current NMR (or Magnetic Resonance Imaging (MRI)) technologies now allows for localization of these signals and relaxation times to examine many more petrophysical information. This petrophysical information includes pore size distributions, capillary pressure, wettability, enhanced oil recovery rate, sweep efficiencies and flooding studies.

This presentation will give an overview of MRI of rocks and discuss the differences between MRI used in the clinical field and that used for porous media like rocks. Example imagines of different rock types taken at 12MHz and 128MHz will be presented. MRI can also be used to directly detect nuclei other than hydrogen such as sodium or carbon. In addition, information such as pore size or flow rates can be localized in the samples creating 4D dataset (x, y, z and flow for example).

Practical Applications For NMR Core Analysis

Authors

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Over the past 30 years, nuclear magnetic resonance (NMR) has become a common tool employed in core analysis laboratories. Routinely, T_2 distributions are employed to derive porosity, T_1 - T_2 maps are used for fluid typing and saturation profiles are used to derive capillary pressure curves of core samples. As these measurements become more common, it is important to understand how these measurements are affected by the NMR parameters employed. For example, the T_2 distribution is derived from the Carr-Purcell-Meiboom-Gill (CPMG) NMR pulse sequence. The CPMG sequence employs many RF pulses (linearly spaced), each of which deposit energy into the sample. This can lead to heating of the sample, which can lead to a reduction in NMR signal, which can lead to inaccuracies in porosity determination and alteration of the pore size distribution. Another example is how NMR data can vary with magnetic field. In recent years, higher field instruments have become more popular because of the faster scan times for a given signal-to-noise ratio. What is not appreciated when employing higher field instruments, is how the T_2 distributions and T_1 - T_2 maps can shift and change as the field is increased. Finally, the quality of T_2 distributions and T_1 - T_2 maps hinge on the inversion of the NMR data from the time domain to the T_2 or T_1 - T_2 domain. The signal-to-noise ratio of the time data plays an important role in this inversion and can greatly affect the appearance of the resulting T_2 distributions and T_1 - T_2 maps. The purpose of this talk is to educate the audience on how the parameters employed can greatly affect the quality of their NMR core analysis results. It is hoped that the audience will learn which pitfalls to avoid when setting the parameters for their NMR core analysis experiments.

Understanding Non-exponential Characteristics of NMR Responses from Unconventional Source Rocks and Impacts to Core Analysis

Z. Harry Xie
Core Laboratories

Abstract

Nuclear magnetic resonance (NMR) has been successfully applied to both well logging and core analysis mainly through the measurements of the spin-spin relaxation time, often known as T_2 or transverse relaxation time, particularly in the cases of conventional reservoir characterization. The T_2 is commonly measured using the Carr-Purcell Meiboom-Gill (CPMG) pulse sequence. Such an experimental method is based on a key factor that the actual NMR signal decay rate (intrinsic T_2) is significantly longer than the decay rate of the free induction decay (T_2^*). In other words, the CPMG echo train technique is mostly applicable to less restricted liquid cases, i.e. the liquids or fluids in large pores or fractures. However, when dealing with unconventional tight rocks, where the spin-spin relaxation has much faster rates of sub-milliseconds range (equivalent to T_2^*), the traditional CPMG method of measuring T_2 faces challenges. Furthermore, it is questionable to assume exponential behaviors of the NMR responses from liquids in pores of nano-meter scales, water in clay and mineral structures, or hydrocarbons in their solid forms. In this work, the general conditions of using the CPMG pulse sequence, together with the exponential function-based inversion techniques are reviewed. Examples of T_2 distributions with various inter-echo spacing times (TE) from unconventional shale samples demonstrate the non-exponential nature of the spin-spin relaxation in source rocks. Simply shortening TE is not a solution. The early-time NMR signals from rigid solids in source rocks provide further evidence of the non-exponential characteristics of the spin-spin relaxation process. The applications of the early-time NMR signals are explored. The experimental observations of this work will lead to more accurate interpretations of NMR data of well logging and core analysis for unconventional reservoirs. It can also have great impact to unconventional core analysis and the development models when applying machine learning techniques.

Investigating Rock Anisotropy Effects on Acoustic and Electrical Properties through Angle-Dependent Measurements

Domenico Crisafulli, Daria Olszowska and Carlos Torres-Verdín

ABSTRACT

Due to various depositional and structural processes rocks frequently exhibit direction-dependent properties. Understanding rock anisotropy and its extent is crucial for accurate formation evaluation. Mislabeling formations as isotropic can result in considerable errors when predicting elastic and electrical properties, leading to substantial disparities between models and real field observations. We integrate angle-dependent ultrasonic reflection coefficient and ellipsometry measurements to attain a comprehensive understanding of the impact of rock anisotropy on estimated acoustic and electrical properties. These methods are applied to synthetic 3D-printed layered samples with varying layer thicknesses. Measurements are conducted both parallel and perpendicular to the bedding plane, providing insights into the effects of anisotropy direction on the estimated properties. Angle-dependent ultrasonic reflection coefficient method yields high-resolution information on sample P- and S-wave velocity and bulk density allowing detection of small-scale changes in rock elastic properties contingent upon measurement location and sample orientation. To validate the results, we benchmarked them against semi-analytical and numerical simulations, and further compared them to the standard acoustic transmission method. Simultaneously, the study explores the impact of rock anisotropy on electrical properties through ellipsometry measurements, revealing anisotropic behavior in refractive index and dielectric properties. Measurements at different angles (orientation) elucidate directional variations in electrical response, providing key information on subsurface permittivity and conductivity.

Validation of ellipsometry results with a standard dielectric probe strengthens the assessment of electrical properties. The experimental data reveal that both methods are influenced by anisotropy direction and scale. Substantial differences emerge when measurements are obtained parallel and perpendicular to the bedding plane. In the former case, the method captures the properties of the layer with minimal deviations (below 5%) compared to the homogeneous material of the same type.

Comparative experimental study of ultrasonic attenuation measurements in H₂ and CO₂-saturated Fontainebleau sandstones

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Abstract

Implementing subsurface storage for greenhouse gases (CO₂) and clean energy carriers (H₂) involves repurposing depleted hydrocarbon and deep saline sandstone reservoirs to address climate change and support clean energy transition. However, fluid injection may affect the storage capacity of reservoirs. Time-lapse seismic surveys and fluid-flow modeling can optimize injection, delineate fluid plume migration, and monitor fluid containment. Field seismic measurements and seismic inversion face challenges quantifying attenuation to constrain poro-viscoelastic models due to insufficient experimental data for H₂ and CO₂ injection. However, attenuation in sands containing H₂ and CO₂ could be studied under laboratory ultrasonic conditions. This study investigates the effects of full H₂ and CO₂ saturation in Fontainebleau sandstones on ultrasonic (1Mhz) velocities and attenuation under increasing effective stress and temperature. The first ultrasonic experiment involved incremental confining pressures (400-2000psi) and pore pressures

(100-900psi). The subsequent ultrasonic measurement involved constant confining stress (1500psi) and pore pressure (500psi) between 22-80°C. The pulse transmission and spectral ratio methods were used to acquire acoustic waveforms (P and S-wave) and measure attenuation coefficients ($1/Q_p$ and $1/Q_s$), respectively. Under applied isostatic stress, the non-linear increase of elastic velocities arises from the closure of compliant pores. Attenuation reduces with effective stress due to diminishing effects of fluid-flow wave relaxation as compliant pores close. Porosity increases attenuation due to fluid-flow dissipative effects from higher volumes of compliant pores. Under applied temperature, the decrease of elastic velocities and increase in attenuation is attributed to reduced fluid density/modulus and wave dissipation from increased molecular collisions, respectively. Generally, H_2 exhibited higher attenuation than CO_2 under applied stress and thermal conditions due to more damped acoustic waves associated with its low molecular weight and high thermal conductivity. The low ultrasonic attenuation is dominated by minimal microscale wave-induced flow of injected weakly viscous gases between the compliant and stiffer intergranular pores.

Methane Gas Hydrate Precipitation Modeling Using Digital Rock Physics

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In-situ solids precipitation in reservoir rock alters key rock properties such as porosity and permeability by plugging the pore space available for fluids to flow. Numerous commercial reservoirs worldwide are subject to the risk of solids precipitation such as asphaltenes, paraffin, salt, and gas hydrate formation. Also, the understanding of CO₂ adsorption, retention, mineralization, and chemical bonding to the surface of the rock is important in the long-term CO₂ trapping mechanism; that is mineralization in carbon capture and storage efforts.

Traditional laboratory measurements are resource-intensive, require intact and clean core samples, and are largely non-repeatable due to experimental artifacts. Reservoir modeling and fluid flow simulations often rely on empirically derived laws to describe the relationship of key rock parameters and average these parameters over the grid cells' structure. Thereby, reservoir simulation generally misses important concepts of fluid flow on the pore scale. Increased availability of computing power enables a new method on the rise called Digital Rock Physics that represents a virtual real rock sample by creating a digital twin. This allows fluid flow simulations incorporating reactions either directly on the segmented pore space of the rock or in a simplification, depending on the required complexity of the simulations and available computational expense.

A previous experimental study (Geranutti, 2020) detected the formation of gas hydrates using multiphysics measurements of core samples from the Bentheimer sandstone outcrop and reservoir fluids from the Wisting field in Norway. Subsequent numerical modeling to reproduce the experimental results (Judmaier, 2021) was able to simulate the hydrate forming reactions and related thermodynamics on the continuum scale, i.e., the laboratory scale. However, it was not possible to quantitatively match resulting permeability effects indirectly calculated via differential pressure measurements. Therefore, we investigate gas hydrate formation on the pore scale using Digital Rock Physics more accurately.

Characterizing the Geomechanical and Geochemical Suitability of Saline Aquifers for Hydrogen Storage

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The development of large-scale energy storage technologies is necessary to integrate renewable energy into the global energy market. Successful large-scale storage of energy will result in a balance between supply and demand and will support the deployment of the low-carbon economy. Some areas for development include seasonal storage of hydrogen, natural gas, and compressed air in the subsurface. Saline aquifers represent promising candidates for subsurface storage, owing to their large storage capacity and abundance. However, for seasonal hydrogen storage, some risks may be associated with storage in these reservoirs. These include mineralogic alteration and the potential of geomechanical failure due to injection/depletion cycles involved in seasonal storage. Field experience with underground storage of hydrogen in porous formations is very limited. Therefore, the aim of this research study is to investigate the impact of detrital and authigenic mineralogy, primary texture (grain size and sorting), and compaction state (a function of burial history as indicated by contact percentage among grains) of siliciclastic reservoirs on their suitability for a safe hydrogen storage.

The research study includes three sandstone outcrop samples exhibiting a significant difference in porosity (or intergranular volume, a function of compaction state), grain size, and mineralogy (both detrital and authigenic). These microstructural features are believed to control the mechanical behavior and ultimate failure of the samples. To develop an unbiased comparison, tested samples have been exposed to the same stress path during laboratory testing. Results show that coarser grained, under compacted, highly feldspathic sands respond to stress cycling by strain weakening, and ultimately failure, illustrating that this type of reservoir, although highly porous and permeable, is not suitable for hydrogen storage. In contrast, more quartzose, more highly compacted, and finer grained sandstones respond to stress cycling by strain hardening, making them promising candidates for hydrogen storage.

Analysis of Seismic Anisotropy in the Kevin Dome, Montana: A CCUS Project—From Core to Seismic

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Abstract:

Characterizing faults, fractures, and in-situ stresses is critical for carbon capture and storage projects to identify areas prone to leakage. Seismic anisotropy provides insights into fracture presence and propagation patterns to identify areas with potential leakage risk. We aim to examine subsurface anisotropy across scales, integrating borehole measurements with 3D seismic data to bridge core and seismic imaging at the Kevin Dome, Montana.

The anisotropy characterization integrates the mechanical laboratory analysis of 43 core samples from the borehole. Additionally, it incorporates a 3D anisotropic shear wireline log and multi-component 3D seismic data. We favored the Schoenberg and Muir theory over the Hudson model, adding layering into the numerical models to honor the effective medium model's implementation and bridge the gap between well-log data and seismic.

Our findings are presented in two parts. First, we present a correlation matrix between density, velocities (P , S_{slow} , and S_{fast}), and anisotropic parameters (ϵ , δ , γ , σ , and η), resulting in a calibrated rock physics template. The second part compares using two synthetic shot gather examples, one incorporating our improved anisotropy template and borehole velocities, and the second calculating anisotropic parameters from migration velocities used during multi-component seismic time migration. Our snapshots show differences in phase velocities.

For the first result, we demonstrated that the density and velocities of the rock are weakly correlated with the anisotropic parameters in the area, suggesting that anisotropy is independent of burial history and rock porosity. Our second results indicate the P-wave group velocity and S-wave group velocity as a function of the dip and azimuth of the fractures, opening the possibility of migrating the seismic data with improved velocity models and including the anisotropic parameters.

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